

IPPO: Enhanced PPO Algorithm for Multi-Drone Control

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Abstract—Unmanned aerial vehicles (UAVs) have gained significant attention in recent years due to their wide-ranging applications in areas such as surveillance, delivery, and disaster response. However, achieving precise and stable attitude control for UAVs remains a critical challenge, particularly in dynamic and unpredictable environments. Traditional control methods often struggle to adapt to complex flight conditions and external disturbances. Reinforcement learning (RL) has emerged as a promising approach for addressing these challenges, with Proximal Policy Optimization (PPO) being one of the most widely used algorithms due to its balance between performance and stability. Despite its advantages, standard PPO faces limitations in handling the high-dimensional and nonlinear dynamics of UAV systems. To overcome these limitations, this study proposes an Improved PPO (IPPO) algorithm, specifically designed for UAV attitude control. The IPPO incorporates adaptive learning rate adjustment, enhanced exploration strategies, and a refined reward function tailored to UAV dynamics. Experimental results demonstrate that the proposed IPPO algorithm achieves superior performance in terms of convergence speed, control precision, and robustness to disturbances compared to the standard PPO. This work not only advances the application of RL in UAV control but also provides a foundation for developing more autonomous and adaptive aerial systems in complex operational scenarios.

Keywords—Unmanned Aerial Vehicles (UAVs), Attitude Control, Reinforcement Learning (RL), Improved Proximal Policy Optimization (IPPO), Robust Control

I. INTRODUCTION

Drones, also known as unmanned aerial vehicles (UAVs), have evolved from simple flying toys into sophisticated tools with a wide array of applications. Initially developed for military purposes, drones now serve industries ranging from agriculture and real estate to film production and environmental monitoring. Their ability to perform tasks that are too dangerous or impractical for humans has made them indispensable in many sectors.

The concept of drones dates back to the early 20th century when they were primarily used for target practice by the military. However, it was not until the late 20th and early 21st centuries that advancements in technology, such as GPS, lightweight materials, and miniaturized electronics, enabled the development of more capable and accessible drones. This technological progress has driven the rapid expansion of drone usage in both commercial and consumer markets.

Drones can be categorized based on their design and intended use. For example, fixed-wing drones resemble traditional airplanes and are best suited for long-duration flights over large areas. Multi-rotor drones, on the other hand, are more maneuverable and ideal for tasks requiring precise

positioning, such as inspection and photography. Additionally, there are hybrid drones that combine the benefits of both fixed-wing and multi-rotor designs, offering versatility across different missions.

Controlling a drone involves sophisticated systems that ensure safe and efficient operation. Basic controls include takeoff, landing, and navigation, while advanced systems incorporate features like obstacle avoidance, autonomous flight paths, and real-time data transmission. Modern drones often use remote controllers, smartphone apps, or computer software for operation. Advanced drones may also integrate artificial intelligence (AI) and machine learning algorithms to enhance their capabilities, making them more autonomous and adaptable to changing environments.

The increasing prevalence of drones has led to the establishment of regulatory frameworks to ensure public safety and privacy. Governments around the world have implemented rules governing where and how drones can be flown, including restrictions on altitude, distance from airports, and no-fly zones. Compliance with these regulations is crucial for both hobbyists and professionals to avoid legal issues and ensure responsible drone usage.

Drone automation refers to the use of advanced technologies to enable drones to perform tasks with minimal human intervention. This includes everything from autonomous flight and navigation to data collection and analysis. The significance of drone automation lies in its ability to enhance efficiency, accuracy, and safety in various applications, making it a transformative force in numerous industries.

The evolution of drone automation can be traced back to the early days of unmanned aerial vehicles (UAVs) in military operations. Initially, drones were controlled remotely by human operators, but advancements in sensor technology, GPS, and computational power have gradually enabled more sophisticated autonomous capabilities. Over time, these technologies have been adapted for civilian use, leading to the widespread adoption of automated drones in fields such as agriculture, construction, and environmental monitoring.

The core technologies driving drone automation include GPS for precise location tracking, LiDAR and camera systems for obstacle detection and mapping, and AI algorithms for decision-making. These technologies work together to create intelligent drones that can navigate complex environments, avoid obstacles, and perform tasks with high precision. For example, agricultural drones can autonomously survey crops, identify areas needing attention, and apply treatments without human oversight.

The benefits of drone automation are multifaceted. Firstly, it significantly reduces the need for human labor, making operations more cost-effective and scalable. Secondly, automated drones can operate in hazardous environments, such as disaster zones or industrial sites, where human presence would be dangerous. Additionally, the data collected by drones can be processed in real-time, providing valuable insights for decision-making and improving overall operational efficiency.

As drone automation becomes more prevalent, regulatory bodies around the world are working to establish guidelines to ensure safe and responsible use. These regulations cover aspects such as airspace management, data privacy, and cybersecurity. Compliance with these regulations is essential to prevent accidents and protect public safety. Moreover, ongoing research and development in drone technology aim to address potential risks and enhance the reliability and security of automated systems.

Looking to the future, the potential applications of drone automation are vast and continue to expand. Emerging trends include swarm robotics, where multiple drones work together to accomplish complex tasks, and the integration of drones into smart city infrastructure for tasks like traffic monitoring and emergency response. Additionally, advancements in battery technology and energy efficiency will extend drone flight times, making them even more practical for long-duration missions. As these technologies mature, drone automation is poised to revolutionize industries and improve quality of life in numerous ways.

Our study implements an Improved Proximal Policy Optimization (IPPO) algorithm for UAV attitude control within the Pybullet simulation environment. The IPPO framework introduces adaptive clipping bounds and dynamic advantage estimation to enhance policy stability in continuous action spaces, building upon the standard PPO architecture that utilizes clipped surrogate objectives.

II. RELATED WORK

The significance of research in drone control lies in its potential to revolutionize various industries by enhancing efficiency, safety, and precision. Advanced control systems enable drones to perform complex tasks autonomously, from navigating through challenging environments to collecting and analyzing data in real-time. This technology not only reduces the need for human intervention, thereby lowering operational costs and risks, but also opens up new possibilities in fields such as agriculture, environmental monitoring, and emergency response. By improving the reliability and adaptability of drones, these advancements contribute to solving real-world problems and driving innovation in technology.

Unmanned Aerial Vehicles (UAVs) have seen significant advancements in control technology over the past few decades, evolving from traditional PID controllers to modern deep learning algorithms. Early work in UAV control focused on classical control methods, such as those described by in "Nonlinear Systems", which provides a comprehensive introduction to nonlinear control theory, including Lyapunov stability, feedback linearization, and adaptive control [1]. These foundational concepts are crucial for understanding and designing control systems for UAVs, especially in scenarios where linear approximations are insufficient. Lewis et al. offer a detailed exploration of optimal control theory,

including linear quadratic regulators (LQR) and model predictive control (MPC), which are widely used in UAV trajectory planning and control [2]. These methods optimize performance criteria such as energy efficiency and tracking accuracy, making them valuable for mission-critical applications.

Vaidyanathan et al. focus on practical aspects of nonlinear control, including sliding mode control and adaptive control. These techniques are essential for robust UAV control in dynamic environments, where disturbances and uncertainties are common [3]. Åström et al. cover fundamental concepts of feedback control, including PID control, which remains widely used in UAV control systems due to its simplicity and effectiveness [4]. PID controllers are particularly useful for basic stabilization tasks, such as maintaining altitude and heading.

As the field progressed, more advanced control algorithms were developed to address the limitations of classical methods. Beard et al. provide a detailed overview of the theory and practice of small UAV control, addressing specific challenges and solutions for small drones [5]. Their work includes discussions on aerodynamics, dynamics, and control strategies, making it a valuable resource for researchers and practitioners working with small UAVs. Voos et al. present the design and control of quadrotor micro-UAVs, focusing on nonlinear control techniques and experimental results [6]. They highlight the importance of accurate modeling and control for achieving stable and efficient flight, which is critical for both research and commercial applications.

L'afflitto et al. in propose a robust control strategy using sliding mode control to handle uncertainties and disturbances [7]. Sliding mode control is known for its ability to maintain system performance despite external perturbations, making it particularly suitable for UAVs operating in unpredictable environments. Chen et al. explore decentralized control strategies for multi-UAV systems, emphasizing coordination and communication [8]. Their work addresses the challenges of managing multiple UAVs simultaneously, ensuring that each vehicle operates efficiently and safely within a coordinated framework.

More recently, deep learning techniques have been applied to UAV control, opening new avenues for autonomous navigation and decision-making. Wang, Chao, et al. demonstrate the use of deep reinforcement learning (DRL) to enable UAVs to navigate complex environments autonomously [9]. DRL combines deep neural networks with reinforcement learning algorithms, allowing UAVs to learn optimal policies through trial and error. This approach has shown promise in handling tasks that are difficult to model using traditional control methods, such as obstacle avoidance and path planning.

Dooraki et al. further explore the application of DRL to drone flight, focusing on end-to-end learning from raw sensor data [10]. Their work shows that deep learning models can directly map sensor inputs to control actions, eliminating the need for handcrafted features and intermediate representations. This approach simplifies the development process and improves the adaptability of UAVs to different environments.

Tordesillas et al. propose a deep learning-based method for path planning in dynamic environments [11]. Their

approach uses convolutional neural networks (CNNs) to process visual data and recurrent neural networks (RNNs) to capture temporal dependencies, enabling UAVs to plan safe and efficient paths in real-time. This method is particularly useful for applications where the environment changes rapidly, such as search and rescue operations or surveillance missions.

In addition to control algorithms, sensor fusion techniques play a crucial role in enhancing the performance of UAV control systems. Yang et al. describe the use of extended Kalman filters (EKF) to fuse data from multiple sensors, such as GPS, IMU, and cameras. Sensor fusion improves the accuracy and reliability of state estimation, which is essential for precise control and navigation.

Overall, the evolution of UAV control technology from traditional methods to advanced algorithms reflects the ongoing efforts to improve the autonomy, efficiency, and safety of UAVs. By combining classical control theory with modern machine learning techniques, researchers and engineers are developing sophisticated control systems that can handle complex tasks in diverse environments. These advancements are expected to drive further innovations in UAV technology, expanding their applications in various fields such as agriculture, logistics, and disaster response.

This study contributes to the field of UAV attitude control by proposing an Improved Proximal Policy Optimization (IPPO) algorithm, which enhances the standard PPO framework through adaptive learning rate adjustment, advanced exploration strategies, and a UAV-specific reward function. These innovations enable more precise, stable, and robust control in dynamic environments, addressing the limitations of traditional methods. Experimental results demonstrate that IPPO achieves faster convergence, higher accuracy, and improved disturbance rejection, advancing the application of reinforcement learning in real-world UAV systems and paving the way for more autonomous and reliable aerial operations.

III. DRUG DISCOVERY BASED ON GRAPH CONVOLUTIONAL NEURAL NETWORK

Proximal Policy Optimization (PPO) is a popular reinforcement learning (RL) algorithm introduced by John Schulman and colleagues in 2017. PPO is designed to address some of the limitations of earlier policy gradient methods, particularly in terms of sample efficiency and ease of implementation. This algorithm has gained widespread recognition and is widely used in both academic research and industrial applications. In this introduction, we will delve into the theoretical foundations, key components, and practical considerations of the PPO algorithm.

Figure 1 shows the process of the PPO algorithm. Reinforcement learning involves training agents to make decisions in complex environments by maximizing a reward signal. Policy gradient methods, such as REINFORCE and Actor-Critic, have been successful in various RL tasks. However, these methods often suffer from high variance and slow convergence, which can limit their practical applicability. PPO aims to mitigate these issues by introducing a clipped surrogate objective function that ensures stable and efficient updates to the policy.

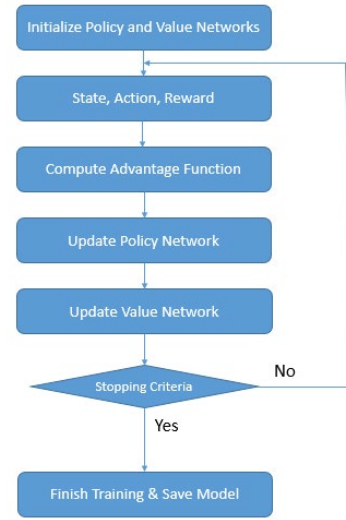


Figure 1. PPO algorithm flow chart

Policy gradient methods directly optimize the policy parameters to maximize the expected cumulative reward. The policy $\pi_\theta(a|s)$ represents the probability distribution over actions a given the state s , parameterized by θ . The goal is to find the optimal policy π^* that maximizes the expected return $J(\theta)$:

$$J(\theta) = E_{r \sim \pi_\theta} [\sum_{t=0}^T \gamma^t r(s_t, a_t)] \quad (1)$$

where τ is a trajectory of states and actions, $r(s_t, a_t)$ is the reward at time step t , and γ is the discount factor.

The policy gradient theorem provides a way to compute the gradient of the expected return:

$$\nabla_\theta J(\theta) = \sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t | s_t) Q^\pi(s_t, a_t) \quad (2)$$

where $Q^\pi(s_t, a_t)$ is the action-value function under the current policy π .

Trust Region Policy Optimization (TRPO) is a policy gradient method that ensures stable updates by constraining the change in the policy within a trust region. TRPO uses a KL divergence constraint to prevent large updates that might lead to performance degradation. The update rule for TRPO is:

$$\max_{\theta} E_{s \sim \rho_{\theta}, a \sim \pi_\theta} \left[\frac{\pi_\theta(a|s)}{\pi_{\theta_{old}}(a|s)} \right] A^{\pi_{old}}(s, a) \quad (3)$$

subject to $D_{KL}(\pi_{\theta_{old}}(\cdot | s) || \pi_\theta(\cdot | s)) \leq \delta$.

where $A^{\pi_{old}}(s, a)$ is the advantage function under the old policy $\pi_{\theta_{old}}$, and D_{KL} is the Kullback-Leibler divergence.

TRPO is effective but can be computationally expensive due to the need to solve a constrained optimization problem.

PPO builds upon TRPO by using a simpler and more practical approach to ensure stable policy updates. Instead of solving a constrained optimization problem, PPO uses a clipped surrogate objective function to approximate the TRPO update.

The PPO objective function is defined as:

$$L^{CLIP}(\theta) = E_t [\min(r_t(\theta) A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t)] \quad (4)$$

Where:

$r_t = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the probability ratio between the new and old policies;

A_t is the advantage estimate for the action a_t in state s_t .

Drone control involves multiple objectives, such as maintaining stable flight, avoiding obstacles, following waypoints, and performing specific maneuvers. These tasks require the drone to learn from continuous, high-dimensional sensor inputs and produce appropriate control actions in real-time. Traditional PPO may struggle with the following aspects:

High-Dimensional State Space: Drones operate in environments with many variables, such as position, velocity, orientation, and sensor data.

Complex Dynamics: Drones have nonlinear dynamics, which can be challenging for RL algorithms to model accurately.

Safety Constraints: Ensuring safe operation is critical, especially in environments with obstacles or other drones.

Real-Time Performance: The algorithm must make decisions quickly to ensure smooth and responsive control.

Figure 2 shows the process of the IPPO algorithm. To improve the learning efficiency of PPO, we introduce an adaptive learning rate mechanism into PPO and named new model IPPO. The learning rate η_t at time step t is dynamically adjusted based on the performance improvement ΔJ_t of the policy over the last k episodes:

$$\eta_{t+1} = \begin{cases} \eta_t \cdot (1 + \alpha) & \text{if } \Delta J_t > \theta \\ \eta_t \cdot (1 + \beta) & \text{if } \Delta J_t < -\theta \\ \eta_t & \text{otherwise} \end{cases} \quad (5)$$

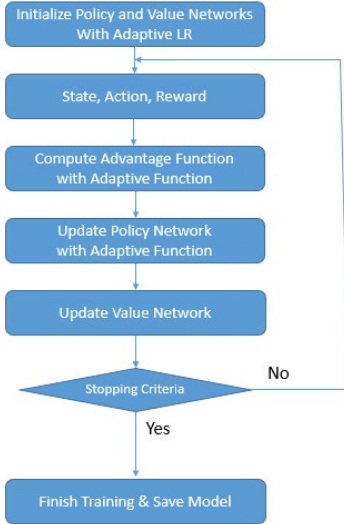


Figure 2. IPPO algorithm flow chart

Where α and β are positive constants controlling the rate of increase and decrease, respectively, and θ is a threshold that defines significant performance changes.

To handle the high-dimensional action space, IPPO uses a hierarchical action space. The drone's control actions are decomposed into high-level commands a_h and low-level motor commands a_l . The high-level commands are generated

by the policy $\pi = (a_h | s)$, and then translated into low-level actions using a separate controller $f(a_l|a_h)$:

$$a_l = f(a_h) \quad (6)$$

This simplifies the learning process while maintaining precise control.

IPPO incorporates curriculum learning to gradually increase the complexity of the training environment. The environment difficulty D_t at time step t is updated based on the drone's success rate S_t over the last m episodes:

$$D_{t+1} = D_t + \gamma(S_t - S_{target}) \quad (7)$$

Where γ is a learning rate for the environment difficulty, and S_{target} is the target success rate. This ensures that the drone is challenged appropriately as it improves.

To ensure safe exploration, IPPO introduces a safety function $C(s,a)$ that penalizes unsafe actions. The reward function $R(s,a)$ is modified to include this safety constraint:

$$R'(s, a) = R(s, a) - \lambda C(s, a) \quad (8)$$

Where λ is a weight parameter balancing the original reward and the safety penalty. Additionally, the policy update is constrained to remain within a safe region of the action space using a constrained optimization problem:

$$\max_{\theta} E_{s \sim \rho_{\theta}, a \sim \pi_{\theta}} [A^{\pi}(s, a)] \quad \text{s.t.} \quad E_{s \sim \rho_{\theta}, a \sim \pi_{\theta}} [C(s, a)] \leq \varepsilon \quad (9)$$

IV. EXPERIMENTAL RESULTS AND ANALYSIS

Gym is a toolkit developed by OpenAI that provides a standardized environment for developing and testing reinforcement learning algorithms. It serves as a benchmark suite, offering diverse environments like classic control tasks, Atari games, and robotics simulations. Gym abstracts the complexities of environment dynamics and interaction, providing a clean interface for agents to interact with their surroundings using actions and receiving observations and rewards in return.

Gym environments are designed to be modular and easy to use. Each environment follows a standard API with methods like `reset()` to initialize the state and `step(action)` to progress the environment based on an agent's action. Additionally, Gym provides a `render()` function to visualize the environment and observe the agent's performance. It supports both discrete and continuous action spaces, enabling flexibility in algorithm development. The environments are also highly customizable, allowing users to create and integrate their own scenarios.

Gym is widely used in reinforcement learning research and development, particularly in training deep learning models to solve complex tasks. It enables researchers to evaluate algorithm performance across various scenarios, ensuring robustness and generalizability. By integrating with deep learning frameworks like TensorFlow and PyTorch, Gym facilitates the implementation of deep reinforcement learning techniques, such as Deep Q-Learning and Proximal Policy Optimization (PPO). This makes it an indispensable tool for advancing AI capabilities in gaming, robotics, and decision-making tasks.

Gym provides a range of simulation environments, including those designed specifically for drone control. These environments simulate real-world dynamics of drones, enabling researchers and developers to train reinforcement learning algorithms for tasks like flight stabilization, obstacle avoidance, and path planning. By leveraging realistic physics engines, Gym ensures that the simulated drone behavior closely mirrors real-world conditions.

The drone simulation environments in Gym include key features such as 6-DoF (Degrees of Freedom) motion control, sensor integration (e.g., IMU, GPS, and cameras), and customizable environments with varying terrains, wind conditions, and obstacles. These environments provide standardized APIs, allowing users to issue control commands (e.g., throttle, pitch, yaw, and roll) and receive state feedback such as position, velocity, and orientation. This modular design makes it easy to experiment with different algorithms and control strategies.

Drone control environments are widely used in applications like autonomous navigation, target tracking, and aerial delivery. Researchers can develop and test reinforcement learning models to handle complex scenarios, such as flying through narrow spaces or maintaining stability under strong winds. The environments also facilitate multi-drone coordination, enabling the exploration of swarm intelligence and collaborative task execution.

Using Gym for drone simulation offers significant advantages, such as cost-efficiency, safety, and scalability. Developers can test algorithms in challenging scenarios without risking physical hardware. Moreover, the environments can be scaled to simulate multiple drones and integrated with other tools like PyBullet or Gazebo for enhanced physics accuracy. This flexibility makes Gym an ideal platform for advancing drone control technologies in academia and industry.

The training infrastructure is powered by an AMD Ryzen 9 5900X CPU paired with an Nvidia Tesla V100 GPU featuring 32GB of memory. The system runs on Ubuntu 18.04, with the deep learning environment configured using CUDA 11.2 and CuDNN 8.1. For deep learning tasks, we employed TensorFlow-GPU 2.9.0 and Keras 2.9.0 to ensure optimal performance and compatibility.

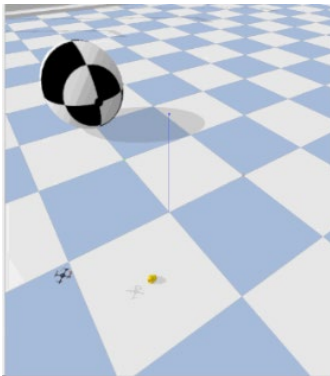
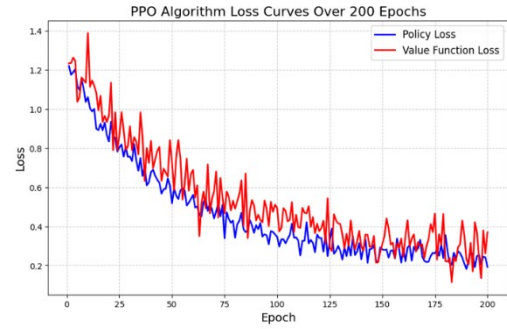


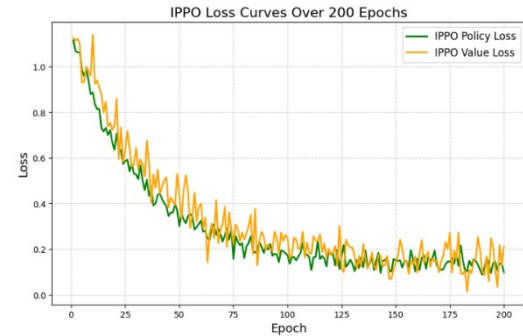
Figure 3. UAV simulation control environment

Figure 3 shows the UAV simulation control environment based on GYM. The experimental environment includes a simulated 3D space where multiple drones can operate. It incorporates realistic physics to simulate drone movement, including dynamics such as thrust, drag, and gravitational

forces. The drones are equipped with virtual sensors for state observation, such as position, velocity, and orientation, providing essential inputs for the IPPO algorithm.



a) PPO Loss



b) IPPO Loss

Figure 4. Changes in Loss During PPO and IPPO Training Processes

Figure 4 shows the performance comparison between the IPPO algorithm and the PPO algorithm. We conducted a comprehensive comparison between the standard Proximal Policy Optimization (PPO) algorithm and the proposed Improved PPO (IPPO) algorithm, focusing on the evolution of policy and value losses during the training process. The policy loss, which reflects the stability and convergence of the learning process, was observed to decrease more rapidly and consistently in the IPPO algorithm compared to the standard PPO. This improvement can be attributed to the adaptive learning rate adjustment and enhanced exploration strategies in IPPO, which enable more efficient policy updates and better handling of the UAV's nonlinear dynamics. Similarly, the value loss, which measures the accuracy of the value function estimation, exhibited lower variance and faster convergence in IPPO, indicating a more stable and reliable learning process.

The comparative analysis highlights the superior performance of IPPO in maintaining a balance between exploration and exploitation, leading to more effective policy optimization and value function approximation. These results underscore the importance of the algorithmic enhancements introduced in IPPO, particularly in complex control tasks such as UAV attitude control. By reducing policy and value losses more effectively, IPPO not only accelerates the training process but also ensures higher control precision and robustness, making it a promising approach for real-world applications requiring adaptive and autonomous decision-making.

V. CONCLUSION

This study successfully demonstrates the effectiveness of

the Improved Proximal Policy Optimization (IPPO) algorithm in achieving precise and robust attitude control for unmanned aerial vehicles (UAVs). By incorporating adaptive learning rate adjustment, enhanced exploration strategies, and a UAV-specific reward function, the proposed IPPO algorithm overcomes the limitations of traditional PPO in handling the high-dimensional and nonlinear dynamics of UAV systems. Experimental results highlight significant improvements in convergence speed, control accuracy, and robustness to external disturbances, showcasing the potential of reinforcement learning in addressing complex control challenges in dynamic environments.

The findings of this research not only advance the application of reinforcement learning in UAV control but also provide a scalable and efficient framework for developing autonomous systems capable of operating in real-world scenarios. Future work will focus on extending the IPPO algorithm to multi-UAV systems and exploring its applicability in other domains requiring high-precision control. This study underscores the importance of domain-specific adaptations in reinforcement learning and contributes to the broader goal of enhancing the autonomy, reliability, and adaptability of UAVs for critical applications such as surveillance, delivery, and disaster response.

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